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Online multivariate changepoint detection: leveraging links with computational geometry

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Abstract

The increasing volume of data streams poses significant computational challenges for detecting changepoints online. Likelihood-based methods are effective, but a naive sequential implementation becomes impractical online due to high computational costs. We develop an online algorithm that exactly calculates the likelihood ratio test for a single changepoint in p-dimensional data streams by leveraging a fascinating connection with computational geometry. This connection straightforwardly allows us to exactly recover sparse likelihood ratio statistics: that is assuming only a subset of the dimensions are changing. Our algorithm is straightforward, fast, and apparently quasi-linear. A dyadic variant of our algorithm is provably quasi-linear, being $\mathcal{O}(n \log(n)^{p+1})$ for n data points and p less than 3, but slower in practice. These algorithms are computationally impractical when p is larger than 5, and we provide an approximate algorithm suitable for such p which is $\mathcal{O}(np \log(n)^{\tilde{p}+1})$, for some user-specified $\tilde{p} \leq 5$. We derive statistical guarantees for the proposed procedures in the Gaussian case, and confirm the good computational and statistical performance, and usefulness, of the algorithms on both empirical data and NBA data.

Keywords: computational geometry, convex hull, dynamic programming, online changepoint detection

1 Introduction

In recent years, many methods have been proposed for detecting one or multiple changepoints offline or online in data streams. The reason for such a keen interest in changepoint detection methods lies in its importance for various real-world applications, including bioinformatics (Olshen et al., 2004; Picard et al., 2005), econometrics (Aue et al., 2006; Bai & Perron, 2003), medicine (Bosc et al., 2003; Staudacher et al., 2005), climate and oceanography (Ducré-Robitaille et al., 2003; Reeves et al., 2007), finance (Andreou & Ghysels, 2002; Fryzlewicz, 2014), autonomous driving (Galceran et al., 2017), entertainment (Radke et al., 2005; Rybach et al., 2009), computer vision (Ranganathan, 2012) or neuroscience (Jewell et al., 2022). Whether this is online (i.e. observations are processed as they become available) or offline (i.e. observations are processed only once the whole data is available) the ever-increasing size of the data stream raises computational challenges. Ideally, a changepoint procedure should be linear or quasi-linear in the size of the data and, for online applications, should also be able to process new observations sequentially, and as

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quickly as they arrive. A common idea for both offline and online parametric (multiple) changepoint model estimation, is to optimize a likelihood function or calculate likelihood ratio statistics (Auger & Lawrence, 1989; Siegmund & Venkatraman, 1995). From a computational perspective this optimization can either be done (a) heuristically using, for example, the Page-CUSUM statistic (Page, 1954) in an online context or Binary Segmentation and its variants (Fryzlewicz, 2014; Scott & Knott, 1974) in an offline context or (b) exactly using dynamic programming techniques as in the Segment Neighbourhood, or Optimal Partitioning algorithms (Auger & Lawrence, 1989; Jackson et al., 2005). Importantly, these later algorithms used with an appropriate penalty enjoy good statistical properties (Garreau & Arlot, 2017; Lavielle & Moulines, 2000; Lebarbier, 2005; Yao, 1984) and have shown good properties in numerous applications (Lai et al., 2005; Liehrmann et al., 2021).

Significant speed-ups are possible by reducing the set of changepoint locations that need to be checked at each iteration of the dynamic programming recursion, a technique commonly known as pruning (Killick et al., 2012; Maidstone et al., 2017; Rigaill, 2015; Romano et al., 2023). Two types of rules have been proposed that prune changepoint locations that will never contribute to the full optimization cost, achieving significant speed-ups without sacrificing optimality. Inequality pruning rules, as in the PELT algorithm (Killick et al., 2012), apply to a wide range of changepoint models but are efficient only if the number of true changepoints increases quickly with the size of the data. Functional pruning, as in the pDPA, FPOP, or FOCuS algorithms (Maidstone et al., 2017; Rigaill, 2015; Romano et al., 2023), applies mostly to univariate models, but is efficient even when the number of true changepoints is small compared to the size of the data. Interestingly, these algorithms are sequential in nature and empirically quasi-linear. This quasi-linearity was recently proven for the online detection of a single changepoint (Romano et al., 2023). The pDPA, FPOP, and FOCuS algorithms rely on a functionalisation of the problem, that is they consider the likelihood as a function of the last segment parameter. A direct extension of these ideas to a higher dimension ($p > 1$) has proven difficult because it involves computing the intersection and subtraction of many convex sets in $\mathbb{R}^p$ (Pishchagina et al., 2024; Runge, 2020).

In this article, we study online models with one changepoint, where the likelihood can be algebraically written as a log-likelihood from the p-variate natural exponential family. This includes, for example, the Gaussian, Poisson, categorical, and Pareto type-I distribution (assuming the minimum value is known) and also the non-parametric e-detectors of Shin et al. (2024) (see Section 5). We extend the functional pruning approach to such p-variate models using a linearization argument embedding all likelihoods into a bigger space of dimension $p + 2$. This idea allows us to demonstrate a link between the functional problem and problems studied in computational geometry.

In particular, we show that the solutions of this extension correspond to identifying the supporting hyperplanes of a particular half-space intersection problem. For models with one changepoint, we demonstrate that this problem further simplifies and is dual to a lower dimensional convex hull problem of dimension $p + 1$. In this latter case, we further prove, assuming independent and identically distributed (i.i.d.) data, that the number of points on this hull at time n is asymptotically $\mathcal{O}(\log^p(n))$. Leveraging this leads to an empirically log-linear per iteration algorithm for calculating the likelihood ratio test statistics for a change at t.

At any time-step of this algorithm, we can exactly compute the maximum likelihood ratio-statistics and maximum sparse likelihood ratio-statistics allowing only s coordinates to change with $s \leq p$. In online supplementary material I, we derive non-asymptotic bounds of these statistics to control the false alarm rate and derive a bound on the expected detection delay. While the computational cost is quasi-linear, our exact algorithms become impractical for large p, due to the $\log^p(n)$ scaling. For such cases, we provide an approximation storing only an order of $\mathcal{O}(p \log^{\tilde{p}}(n))$ candidates on the hull, where $\tilde{p} \leq p$, chosen by the user, represents the dimension of a projective linear space.

The article has the following structure. Section 2 describes our maximum likelihood framework. Section 3 presents the mathematical formulation of the functional pruning problem and how it is related to a half-space intersection problem and a convex hull problem. In Section 4, we develop and introduce our proposed method, called MdFOCuS, together with an approximation that scales to larger values of p. An important feature of both algorithms is that they calculate sets of putative changepoint locations. Given such a set it is straightforward to maximize the test statistic of interest, including under constraints on the number of components that change, or allowing

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different models for different components. We provide statistical guarantees of the resulting test in the Gaussian setting and compare MdFOCuS with competitor algorithms in terms of statistical and computational efficiency—showing its increased accuracy when the pre-change parameters are unknown. In Section 5, we apply our methodology to track the performances of an NBA team and discuss how our theory could accelerate the recently proposed non-parametric e-detectors (Shin et al., 2024). All proofs, and links to code for this article, are given in the online supplementary material.

2 Maximum likelihood methods for online single changepoint detection

2.1 Problem set-up

We consider observations $\{y_t\}_{t=1,2,\ldots}$ in $\mathbb{R}^{p'}$ and wish to detect the possible presence of a changepoint and estimate its location. If there is a change, we denote the time of the change as τ. We will consider likelihood-based approach that postulates a model for the data in the segments before and after the changepoint. Our model will be that the data point y_t is i.i.d. from a distribution with density $f_Y(y_t \mid \theta_1)$ before the change and $f_Y(y_t \mid \theta_2)$ after the change.

At time n, twice the log-likelihood for the time series $\{y_t\}_{t=1,\ldots,n}$, in terms of the pre-changepoint parameter, θ_1 in $\mathbb{R}^{p''}$, the post-changepoint parameter, θ_2 in $\mathbb{R}^{p''}$, and the location of a change, τ, is

$$\ell_{\tau,n}(\theta_1, \theta_2) = 2\left(\sum_{t=1}^{\tau} \log f_Y(y_t \mid \theta_1) + \sum_{t=\tau+1}^{n} \log f_Y(y_t \mid \theta_2)\right). \tag{1}$$

Depending on the application one may assume that the pre-changepoint parameter θ_1 is either known or unknown. In either case, we can get the log-likelihood for a changepoint at τ by maximizing this equation over either the unknown parameter, θ_2, or parameters, θ_1 and θ_2. In practice, we do not know the changepoint location, so we maximize this log-likelihood over τ. We can define the maximum likelihood estimator for τ, for the pre-changepoint parameter being either known or unknown respectively, as

$$\hat{\tau}(\theta_1) = \arg\max_{\tau}\left(\max_{\theta_2} \ell_{\tau,n}(\theta_1, \theta_2)\right),$$
$$\hat{\tau}(.) = \arg\max_{\tau}\left(\max_{\theta_1,\theta_2} \ell_{\tau,n}(\theta_1, \theta_2)\right).$$

We then obtain a generalized likelihood ratio (GLR) statistic (Siegmund & Venkatraman, 1995) for a change τ by comparing it with twice the log-likelihood for a model with no change. That is $2\sum_{t=1}^{n} \log f_Y(y_t \mid \theta_1)$ if θ_1 is known and $\max_{\theta_1} 2\sum_{t=1}^{n} \log f_Y(y_t \mid \theta_1)$ if it is not. Often it is appropriate (see, for example, Chen et al., 2022) to consider a sparse version of the likelihood that allows only s coordinates to jump, that is adding the constraint $s = \|\theta_1 - \theta_2\|_0$, where $\|\mathbf{v}\|_0$ is equal to the number of non-zero entries of $\mathbf{v}$. The estimator of τ in this case, assuming the pre-change mean is known, is

$$\hat{\tau}^s(\theta_1) = \arg\max_{\tau}\left(\max_{\substack{\theta_2 \\ \|\theta_1 - \theta_2\|_0 = s}} \ell_{\tau,n}(\theta_1, \theta_2)\right), \tag{2}$$

with $\hat{\tau}^s(.)$ defined analogously for the case where the pre-change mean is unknown. Clearly, $\hat{\tau}^p(\theta_1) = \hat{\tau}(\theta_1)$ and $\hat{\tau}^p(.) = \hat{\tau}(.)$.

Offline, optimizing Equation (1) is simple. Indeed, assuming we have access to all the data, it suffices to compute, for every possible changepoint location from 1 to $n-1$, the likelihood (which can be done efficiently using summary statistics) and then pick the best value. Online, the problem is substantially more difficult as one needs to recompute the maximum for every new observation. Applying the previous offline strategy for every new observation is inefficient as it takes a

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computational cost of order n to process the nth observation. It is infeasible for an online application, where n grows unbounded, and for this reason, various heuristic strategies have been proposed (see Wang & Xie, 2024, for a recent review). From a computational perspective, we roughly identify two main types of heuristics. The first idea is to restrict the set of possible (θ_1, θ_2) pairs to be discrete. The simplest case is to consider just one pair of (θ_1, θ_2) as in the Page CUSUM strategy (Page, 1954), but one typically considers a grid of values (Chen et al., 2022). The second idea is to restrict the set of changepoint locations one considers. For example, to consider a grid of window lengths $w_1, \ldots, w_M$ and at time n compute the GLR statistic only for changes at $\tau = n - w_1, \ldots, n - w_M$ (Eiauer & Hackl, 1978; Meier et al., 2021). But this naturally introduces a trade-off between computational efficiency and statistical power: the larger the grid of windows, the larger the computational overhead (Wang & Xie, 2024).

Recently, the FOCuS algorithm has been introduced that calculates the GLR test exactly but with a per-iteration cost that increases only logarithmically with the amount of data (Romano et al., 2023; Ward et al., 2025)—however, this algorithm is only applicable for detecting changes in a univariate feature of a data stream. A key insight from the FOCuS algorithm is that to maximize the likelihood, one only needs to consider changes that correspond to points on a particular convex hull in the plane. This link to geometrical features of the data is one that we leverage to develop algorithms for detecting changes in multivariate features.

2.2 Detection of a single change in the parameter of a distribution in the exponential family

We briefly introduce some notation. We use $\mathbf{1}_p$ to denote a vector of length p in which all elements are set to one. For any vector $\mu = (\mu^1, \ldots, \mu^p)$ in $\mathbb{R}^p$, we define its q-norm as $\|\mu\|_q := (\sum_{k=1}^p |\mu^k|^q)^{1/q}$, where $q \geq 1$ in $\mathbb{R}$. For two vectors, $\mu_1 = (\mu_1^1, \ldots, \mu_1^p)$ and $\mu_2 = (\mu_2^1, \ldots, \mu_2^p)$ in $\mathbb{R}^p$, we define their scalar product as $\langle \mu_1, \mu_2 \rangle = \sum_{k=1}^p \mu_1^k \mu_2^k$. The symbol $|\cdot|$ signifies the cardinality of $\cdot$.

We are now ready to describe our model. We assume that the density $f_Y(y_t \mid \theta_k)$, for observation y_t in segment $k \in \{1, 2\}$ which has segment-specific parameter θ_k, can be written in the form

$$f_Y(y_t \mid \theta_k) = \exp\left\{-\frac{1}{2}\left(A'(r(\theta_k)) - 2\langle s(y_t), r(\theta_k)\rangle + B'(s(y_t))\right)\right\}, \tag{3}$$

where s and r are functions mapping respectively the observations, y_t, and the parameters, θ_k, to $\mathbb{R}^p$, A' is some convex function from $\mathbb{R}^p$ to $\mathbb{R}$, and B' from $\mathbb{R}^p$ to $\mathbb{R}$. Defining the natural parameter η_k as $\eta_k = r(\theta_k)$, and x_t as $x_t = s(y_t)$ we rewrite this density as a function of η_k and x_t:

$$f_X(x_t \mid \eta_k) = \exp\left\{-\frac{1}{2}\left(A'(\eta_k) - 2\langle x_t, \eta_k\rangle + B'(x_t)\right)\right\}. \tag{4}$$

This class encompasses many common exponential family models, see Table 4 in the online supplementary material. There, we also show how a change in mean and variance of a Gaussian signal in $\mathbb{R}$ can be expressed in this form.

We can then define the log-likelihood of our (transformed) data $x_1, \ldots, x_t$, under a model with a change at τ in terms of η_1 and η_2 as

$$\ell_{\tau,n}(\eta_1, \eta_2) = -\sum_{t=1}^{\tau}\left[A'(\eta_1) - 2\langle x_t, \eta_1\rangle + B'(x_t)\right] - \sum_{t=\tau+1}^{n}\left[A'(\eta_2) - 2\langle x_t, \eta_2\rangle + B'(x_t)\right]. \tag{5}$$

The log-likelihood estimator for τ, for the pre-change parameter being either known or unknown respectively, is

$$\hat{\tau}(\eta_1) = \arg\max_{\tau}\left(\max_{\eta_2} \ell_{\tau,n}(\eta_1, \eta_2)\right), \quad \text{or } \hat{\tau}(.) = \arg\max_{\tau}\left(\max_{\eta_1,\eta_2} \ell_{\tau,n}(\eta_1, \eta_2)\right).$$

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The GLR statistic for a change τ is obtained by comparing with $\sum_{t=1}^{n}[A'(\eta_1) - 2\langle x_t, \eta_1\rangle + B'(x_t)]$ if η_1 is known and with $\max_{\eta_1}\sum_{t=1}^{n}[A'(\eta_1) - 2\langle x_t, \eta_1\rangle + B'(x_t)]$ if it is not.

For any fixed η_1 and any η_2, the addition of $\sum_{t=1}^{n}[A'(\eta_2) - 2\langle x_t, \eta_2\rangle + B'(x_t)]$ to (5) does not change $\arg\max_\tau \ell_{\tau,n}(\eta_1, \eta_2)$. Moreover, this addition eliminates the dependence on n. Hence, when searching for the maximum likelihood change, we can redefine $\ell_{\tau,n}(\eta_1, \eta_2)$ as follows:

$$\ell_\tau(\eta_1, \eta_2) = \tau\big[A'(\eta_2) - A'(\eta_1)\big] - 2\left\langle \sum_{t=1}^{\tau} x_t, \eta_2 - \eta_1 \right\rangle, \tag{6}$$

and be sure that the optimal change optimizes $\ell_\tau(\eta_1, \eta_2)$ at least for a pair (η_1, η_2). Sometimes it might make sense to regularize the parameters. This can be done by adding a function of the parameter $\Omega(\eta_1, \eta_2)$, such as $\Omega(\eta_1, \eta_2) = \lambda(\|\eta_1\|_1 + \|\eta_2\|_1)$ for some $\lambda > 0$, to (5). We will not study this further, but by the previous addition trick, any such regularization term could be subtracted, and the rest of our argument would hold.

Throughout, we will primarily focus on the case where η_1 is known. Our results for the case where it is unknown are derived as a by-product: considering all possible η_1 simultaneously and realizing they all lead to the same geometrical problem. Considering η_1 to be fixed we thus introduce the following definitions: $\mu = \eta_2 - \eta_1$, $A(\mu) = A'(\mu + \eta_1) - A'(\eta_1)$, and rewrite the function $\ell_\tau(\eta_1, \eta_2)$ as a function of μ:

$$\ell_\tau(\mu) = \tau A(\mu) - 2\left\langle \sum_{t=1}^{\tau} x_t, \mu \right\rangle. \tag{7}$$

3 Linear relaxation of the functional pruning problem

3.1 Mathematical formulation

Functional pruning ideas give the potential of an efficient approach to recursively calculate the GLR statistic for a change at time $n+1$ given calculations at time n (Romano et al., 2023). The idea is to restrict the set of possible changepoints prior to $n+1$ that we need to consider. Working out which changepoint locations we need to consider is obtained by viewing the GLR statistic as a function of the change parameter $\mu = \eta_2 - \eta_1$, and finding the set of changepoint locations which contribute to this, i.e. for which the GLR statistic is maximum for some value of μ.

In more detail, at step n of a functional pruning algorithm, we consider a list of possible changepoints τ of the data from $x_1, x_2 \ldots$ to x_n. Each changepoint is represented by a function f_τ of the last segment parameter μ and a set of μ values for which it is maximizing the likelihood: $Set_\tau = \{\mu \mid \forall\, \tau' \neq \tau, f_\tau(\mu) > f_{\tau'}(\mu)\}$. To proceed to the next data point $n+1$, the update is informally done as follows.

1. We add n to the list of changepoints, with this represented by a function f_n.
2. We compare all other functions to this new function and accordingly restrict the set of μ for which they are optimal.
3. We discard all changepoints whose set Set_τ is empty.

The left column of Figure 1 provides a graphical representation of these steps. The pruning of the third step is valid because, whatever the data points after $n+1$ might be, these functions (or equivalently changepoints) can never maximize the log-likelihood. Formally, define functions f_τ, for τ in some set $\mathcal{T}$, such that

$$f_\tau(\mu) = a_\tau A(\mu) - 2\langle b_\tau, \mu\rangle, \tag{8}$$

where a_τ is in $\mathbb{R}$, and b_τ is in $\mathbb{R}^p$ and A is a convex function from $\mathcal{D}_A$, a subset of $\mathbb{R}^p$. We define $Im(A)$ as the image of the function A. Function f_τ corresponds to the log-likelihood for a segmentation

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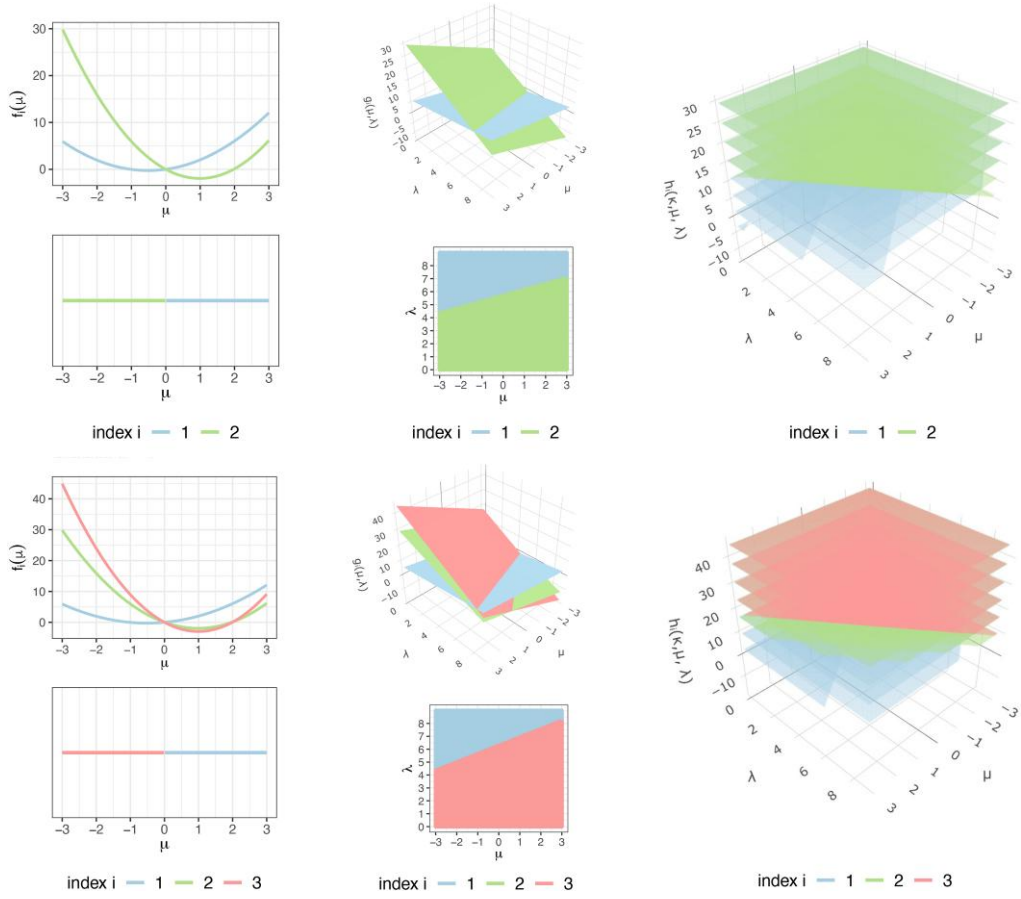


Figure 1. Example of the functional pruning problem and linearized functional problem over times: f_i and $\{\max_i \{f_i(\mu)\}\}$ functions (left); g_i and $\{\max_i \{g_i(\mu, \lambda)\}\}$ functions (middle); the level surfaces of h_i functions, where each surface is for a fixed κ value (right) for changepoint candidates $i \in \mathcal{T}$. We simulate time series $\{x_t\}_{t=1,2,\ldots} \sim \mathcal{N}(0, 1)$ and consider two time-steps, $n = 3$ (top two rows) and $n = 4$ (bottom two rows). We define the coefficients a_i as i, b_i as $\sum_{t=1}^{i} x_t$. The change $i = 2$ associated with quadratics f_2 and surface g_2 is pruned at $n = 4$.

with a change at τ and with parameter μ. This function will depend on μ through the log-likelihood contribution of the data in the segment after τ, with these data defining the co-efficients a_τ and b_τ. We then define

$$F_{\mathcal{T}}(\mu) = \max_{\tau \in \mathcal{T}} \{f_\tau(\mu)\},$$

and our goal is to recover all the indices τ maximizing $F_{\mathcal{T}}$ for at least one value of μ. Define:

$$\mathcal{F}_{\mathcal{T}} = \{\tau \,|\, \exists \mu \in \mathcal{D}_A \subset \mathbb{R}^p, \forall \tau' \in \mathcal{T} \text{ with } \tau' \neq \tau : f_\tau(\mu) > f_{\tau'}(\mu)\}. \tag{9}$$

This is defined using a strict inequality. As function A is strictly convex, (and assuming that for all $\tau \neq \tau'$ we cannot have simultaneously $a_\tau = a_{\tau'}$ and $b_\tau = b_{\tau'}$) equality can be obtained only over a set of zero measure, and can be ignored.

3.2 Linearization of the functional pruning problem

The size of the set $\mathcal{F}_{\mathcal{T}}$ depends on the nature of the function A. In essence, to consider any possible function A, we introduce a new parameter λ to replace $A(\mu)$. This way we get a problem that does not depend on the form of A and, as we will see, is easier to solve, albeit at the cost of an additional

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dimension. To be specific, for any function $\mu \mapsto f_\tau(\mu)$ we associate an (augmented) function $(\lambda, \mu) \mapsto g_\tau(\lambda, \mu)$ defined as:

$$g_\tau(\lambda, \mu) = a_\tau \lambda - 2\langle b_\tau, \mu \rangle.$$

As for the f_τ, we define the maximum $G_\mathcal{T}(\lambda, \mu)$ as

$$G_\mathcal{T}(\lambda, \mu) = \max_{\tau \in \mathcal{T}} \{g_\tau(\lambda, \mu)\}.$$

Again our goal will be to recover the index τ strictly maximizing $G_\mathcal{T}$:

$$\mathcal{G}_\mathcal{T} = \{\tau \,|\, \exists (\lambda, \mu) \in Im(A) \times \mathcal{D}_A, \forall \tau' \in \mathcal{T} \text{ with } \tau' \neq \tau : g_\tau(\lambda, \mu) > g_{\tau'}(\lambda, \mu)\}.$$

In the previous definition, the parameter λ is constrained to be in the image of the function A. The values attained by f_τ are also attained by g_τ on a parametric curve of the (λ, μ) space given by equation $\lambda = A(\mu)$, so that $f_\tau(\mu) = g_\tau(A(\mu), \mu)$. This trivially leads to the following result that the set of τ strictly maximizing $\mathcal{G}_\mathcal{T}$ must contain the set strictly maximizing $\mathcal{F}_\mathcal{T}$.

Theorem 1 $\mathcal{F}_\mathcal{T} \subseteq \mathcal{G}_\mathcal{T}$.

Without any further assumptions on the function A, the set $\mathcal{G}_\mathcal{T}$ could be much larger than $\mathcal{F}_\mathcal{T}$. The following theorem demonstrates that equality holds for an important class of functions $A(\mu)$.

Theorem 2 If $A(\mu) = \|\mu\|^q$ with $q > 1$, then $\mathcal{F}_\mathcal{T} = \mathcal{G}_\mathcal{T}$.

In particular, with $q = 2$ we get the Gaussian loss function with $\eta_1 = 0$. Assuming $\eta_1 = 0$ is not restrictive because if η_1 is not 0 we can consider the translated data $(x_t - \eta_1)$. A more general result, with milder conditions for function A, is proposed in online supplementary material D. One sufficient condition consists of the strong convexity of the function A with $A(0) = 0$.

3.3 Half-space intersections and functional pruning

All g_τ functions are linear in (λ, μ). Considering yet another variable κ we can associate to every g_τ a function h_τ and a half-space H_τ of $\mathbb{R}^{p+2}$:

$$H_\tau = \{\kappa, \lambda, \mu \,|\, h_\tau(\kappa, \lambda, \mu) = a_\tau \lambda - 2\langle b_\tau, \mu \rangle - \kappa \leq 0\}.$$

From here, we straightforwardly get that our linearized functional problem is included in a half-space intersection problem. This result is formalized in the following theorem.

Theorem 3 If τ is in $\mathcal{G}_\mathcal{T}$ then

1. there exists (κ, λ, μ) such that $h_\tau(\kappa, \lambda, \mu) = 0$ and for all $\tau' \neq \tau$, $h_{\tau'}(\kappa, \lambda, \mu) < 0$;
2. $h_\tau(\kappa, \lambda, \mu) = 0$ defines a supporting hyperplane of the set $\cap_\tau H_\tau$.

The set $\cap_\tau H_\tau$ defines an unbounded polyhedron as taking any $\kappa \geq \max_\tau \{c_\tau\}$, $\lambda = 0$, and $\mu = 0$ we solve all inequalities. In Figure 1, we illustrate our linearized functional problem.

Computing the intersection of $|\mathcal{T}|$ half-spaces is a well-studied problem in computational geometry. It is closely related to the convex hull problem by projective duality (Avis & Bremner, 1995), and several algorithms have been proposed (see for example Barber et al., 1996; Brondsted, 1983; O'Rourke, 1998; Preparata & Muller, 1979). See Seidel (2017) for a recent short introduction to this topic.

3.4 Projective duality

The following theorem states that in our case the half-space intersection problem in dimension $p + 2$ is dual to a convex hull problem in dimension $p + 1$.

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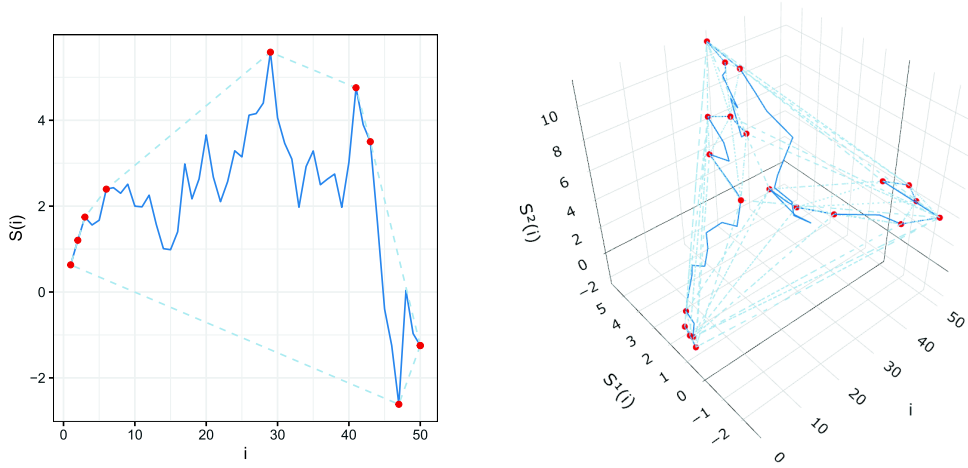


Figure 2. Plot of $S(i) = \sum_{t=1}^{i} x_t$ as a function of i (solid line) and the convex hull of $\{(a_i = i,\ b_i = S(i))\}_{i\in\{1,\ldots,n-1\}}$ (dashed line) for dimensions $p = 1$ (left) and 2 (right). The vertices of the convex hull $\tau \in \mathcal{T}$ are bullet points. We simulated time series $\{x_t\}_{t\in\{1,\ldots,n\}}$ with $n = 51$ data points with $x_t \sim \mathcal{N}_p(0,\ I_p)$.

Theorem 4 An index τ being in $\mathcal{G}_{\mathcal{T}}$ is equivalent to the point $(a_\tau,\ b_\tau)$ in $\mathbb{R}^{p+1}$ being a vertex of the convex hull of $\{(a_\tau,\ b_\tau)\}_{\tau\in\mathcal{T}}$.

We illustrate the points on the convex hull of Theorem 4 for $p = 1$ and $p = 2$ in Figure 2. As will be seen in Section 4, this is relevant for the online detection of a single changepoint, see (7). Importantly, computing the convex hull of $|\mathcal{T}|$ points is a well-studied problem in computational geometry. Over the years many algorithms have been developed (Barber et al., 1996; Chand & Kapur, 1970; Kenwright, 2023; Miller & Stout, 1988; Preparata & Hong, 1977). In the rest of this article, we will use the `QuickHull` algorithm of Barber et al. (1996) and its implementation in the `qhull` library http://www.qhull.org/. The `QuickHull` algorithm works in any dimension and is reasonably fast for our application up to $p = 5$.

4 On the detection of a single changepoint online

4.1 From the functional description to a convex hull

We now apply the results of Section 3 to the problem of detecting a single changepoint described in Section 2.2, to develop an efficient online algorithm. To do this, it is helpful to define a one-to-one mapping P between any changepoint τ and a point in dimension $p + 1$:

$$P(\tau) = \left(\tau,\ \sum_{t=1}^{\tau} x_t\right). \tag{10}$$

The following theorem, based on Theorem 4 essentially states that the changepoint maximizing the likelihood (5) for some value η_1 and η_2 corresponds to a point on a particular convex hull.

Theorem 5 Assume the log-likelihood, $\ell_{\tau,n}(\eta_1,\ \eta_2)$ of a change at τ in $\{1,\ \ldots,\ n-1\}$ can be written as in (5). Then, for any η_1 and $s \le p$ taking $\hat{\tau} = \hat{\tau}^s(\eta_1)$ or $\hat{\tau}^s(.)$ we have that $P(\hat{\tau})$ is on the convex hull of $\{P(\tau)\}_{\tau\in\{1,\ldots,n-1\}}$.

Remark 1 This result also extends to other test statistics that maximize the sparse log-likelihood for some s. This includes statistics that threshold the contribution from each component of the time series (see Section I.1 in online supplementary material).

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Theorem 5 is coherent with the notion of pruning as in the FOCuS algorithm (Romano et al., 2023). If $P(\tau_0)$ is strictly inside the hull, i.e. not on the boundary of the hull $\{P(\tau)\}_{\tau\in\{1,\dots,n-1\}}$ at time n, then it is inside the hull at any future time $n' \geq n$. Thus, based on Theorem 5, the likelihood associated with the change at time τ_0 will never be optimal after time n.

4.2 Bound on the number of changepoint candidates

From a computational perspective, Theorem 5 is useful only if the set of points on the convex hull is significantly smaller than n. We now use recent results (Kabluchko et al., 2017) to bound the expected number of faces and vertices of the convex hull $\{P(\tau)\}_{\tau\in\{1,\dots,n-1\}}$, assuming that the components of x_τ, x_τ^d (with $d = 1, \dots, p$), are i.i.d. with a continuous distribution. In turn, using Theorem 4, this gives us an upper bound on the expected number of changepoints that can maximize the likelihood function (5) for some value of η_1 and η_2. This bound will be useful to derive guarantees on the computational complexity of the procedure (see Algorithms 1 and 3).

Theorem 6 Assuming all x_τ^d (with $d = 1, \dots, p$) are i.i.d. with a continuous distribution, then the expected number of faces (denoted as U_n^p) and vertices (denoted as V_n^p) of the convex hull of $\{P(\tau)\}_{\tau\in\{1,\dots,n-1\}}$ are the following:

$$\mathbb{E}(U_n^p) = \frac{2p!}{(n-1)!}\begin{bmatrix} n \\ p+1 \end{bmatrix}, \quad \mathbb{E}(V_n^p) = \frac{2}{(n-1)!}\sum_{l=0}^{\infty}\begin{bmatrix} n \\ p+1-2l \end{bmatrix},$$

where $\begin{bmatrix} n \\ m \end{bmatrix}$ are the Stirling numbers of the first kind (Adamchik, 1997).

Corollary 1 We have the following equivalents for the expected numbers of faces U_n^p and vertices V_n^p:

$$\mathbb{E}(U_n^p) \sim 2\big(\log(n)\big)^p, \quad \mathbb{E}(V_n^p) \sim \frac{2}{p!}\big(\log(n)\big)^p, \quad n \to \infty.$$

The proof of Corollary 1 follows from Remark 1.4 of Kabluchko et al. (2017). The theorem and corollary provide an expected upper limit on the number of points, or candidate locations for changepoints, that need to be evaluated to compute the CUSUM likelihood ratio test (5) exactly without resorting to approximations. This limit increases as $\mathcal{O}(\log(n)^p)$ and so will give small bounds on the number of candidate changepoints for small p. We empirically illustrate the validity of Theorem 6 for i.i.d. Gaussian and Poisson data in online supplementary material B.5 and F.1.

4.3 MdFOCuS: a simple online algorithm

In this section, we propose an algorithm to compute the GLR statistics at every time step n for a known and unknown pre-change parameter η_1. In both cases, based on Theorem 5 we only need to consider the changepoints τ such that $P(\tau)$ is on the hull of $\{P(\tau')\}_{\tau'\in\{1,\dots,n-1\}}$.

Let us call $\mathcal{T}_n$ the set of index τ such that $P(\tau) = (\tau, \sum_{t=1}^{\tau} x_t)$ is on the convex hull of all $\{P(\tau)\}_{\tau\in\{1,\dots,n-1\}}$. Restating Theorem 4, the changepoint $\hat{\tau}$ maximizing the likelihood (5) is in $\mathcal{T}_n$. Hence, at step n, we only need to compute the likelihood (and likelihood ratio statistics) of changepoints τ in $\mathcal{T}_n$. Furthermore, as already explained after Theorem 5 we can prune all τ that are not in $\mathcal{T}_n$ as they will never be in $\mathcal{T}_{n'}$ for any $n' \geq n$.

Computing the convex hull from scratch at every time step n is inefficient: it requires at least $\mathcal{O}(n)$ operations and leads to an overall complexity at least quadratic in n. Ideally, we would like to update the hull using an efficient online or dynamic convex hull algorithm as the one in Clarkson and Shor (1989). There exist implementations of such an algorithm in dimensions $p = 2$ (Melkman, 1987) and $p = 3$ (Hert & Schirra, 2023). However, for simplicity, here we rely on an iterative use of the offline `QuickHull` algorithm (Barber et al., 1996) that also works for larger dimensions and is available in the `geometry` `R` package (Habel et al., 2023).

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Algorithm 1 MdFOCuS algorithm

1: **Inputs:** $\{x_t\}_{t=1,2,\ldots}$ p-variate independent time series; *thrs* threshold; η_1 pre-change parameter (if known); $maxSize > p+1$, $\alpha \geq 1$, $\beta \geq 0$, pruning parameters.
2: **Outputs:** stopping time n and maximum likelihood change : $\hat{\tau}(.)$ or $\hat{\tau}(\eta_1)$
3: $GLR \leftarrow -\infty, \quad \mathcal{T} \leftarrow \emptyset, \quad n \leftarrow \infty$
4: **while** $(GLR < thrs)$ **do**
5: $\quad n \leftarrow n+1$
6: $\quad \mathcal{T} \leftarrow \mathcal{T} \cup \{n-1\}$
7: $\quad \hat{m} \leftarrow \max_{\tau \in \mathcal{T}} \{\max_{\eta_1,\eta_2} \ell_{\tau,n}(\eta_1,\eta_2)\}$ ▷ η_1 is fixed if pre-change known
8: $\quad GLR \leftarrow \hat{m} - \max_{\eta_1,\eta_2} \ell_{n,n}(\eta_1,\eta_2)$ ▷ η_1 is fixed if pre-change known
9: $\quad$ **if** $(|\mathcal{T}| > \texttt{maxSize})$ **then**
10: $\quad\quad \mathcal{T} \leftarrow$ QuickHull$(\{P(\tau)\}_{\tau \in \mathcal{T}})$ ▷ Find the indices on the hull of all $P(\tau)$
11: $\quad\quad \texttt{maxSize} \leftarrow \lfloor \alpha |\mathcal{T}| + \beta \rfloor$
12: $\quad$ **end if**
13: **end while**
14: **return** n and GLR

In detail, for any $n' > n$ we have $\mathcal{T}_{n'} \subset \mathcal{T}_n \cup \{n, \ldots, n'-1\}$. Therefore, if we have already computed $\mathcal{T}_n$, we only apply `QuickHull` to the points $P(\tau)$ with τ in $\mathcal{T}_n \cup \{n, \ldots, n'-1\}$. A first idea would be to apply the `QuickHull` algorithm at every time step n, $n+1$, ..., that is compute $\mathcal{T}_n$ as the hull of all $P(\tau)$ with τ in $\mathcal{T}_{n-1} \cup \{n\}$. In practice, we discovered that running the `QuickHull` algorithm intermittently, rather than at every iteration, resulted in faster run times. Specifically, we maintain a set of linearly increasing indices, denoted as $\mathcal{T}'_n$, and only periodically execute `QuickHull` to reconstruct the true $\mathcal{T}_n$. More precisely, to decide when to run `QuickHull`, we update a maximum size variable for $\mathcal{T}'_n$ (denoted `maxSize`). Then, at step $n+1$:

- if $|\mathcal{T}'_n|$, is smaller than `maxSize` we set $\mathcal{T}'_{n+1}$ to $\mathcal{T}'_n \cup \{n\}$,
- otherwise, we compute $\mathcal{T}'_{n+1}$ as the convex hull of all $P(\tau)$ with τ in $\mathcal{T}'_n \cup \{n\}$.

In the latter case, we also update `maxSize` to $\texttt{maxSize} = \lfloor \alpha |\mathcal{T}'_{n+1}| + \beta \rfloor$, with $\alpha \geq 1$ and $\beta \geq 0$. It is important to note that, in this way, we always retain a few more possible changepoints than necessary: $\mathcal{T}_{n+1} \subseteq \mathcal{T}'_{n+1}$.

If $\beta = 1$ and $\alpha = 1$, we run the `QuickHull` algorithm at every time step. Larger values of α mean we run it less often, but at the cost of having a longer list of indices in $\mathcal{T}'_n$. We empirically evaluated different values of α with $\beta = 1$, and found $\alpha = 2$ gave good practical performance; see online supplementary material F.2.

A formal description of our procedure, called MdFOCuS is presented in Algorithm 1. The algorithm covers both known and unknown pre-change parameters η_1, with the only difference being the calculation of the likelihood (lines 7 and 8). Therefore, the two variants store the same number of candidate changepoints and have the same time and memory complexities. In detail, computing the full likelihood ratio statistics for any particular changepoint has a time complexity of $\mathcal{O}(p)$, and therefore lines 7 and 8 in Algorithm 1 have a complexity of $\mathcal{O}(|\mathcal{T}|p)$. Recovering the maximum has complexity $\mathcal{O}(|\mathcal{T}|)$. Overall, this is $\mathcal{O}(|\mathcal{T}|p)$.

A link to a simple `R` implementation of Algorithm 1 for a change in the mean of p-dimensional Gaussian signal can be found in the online supplementary material. Our code is based on the `convhulln` function implementing the `QuickHull` algorithm in the `geometry R` package, and for $p = 3$ it takes 2 min to process $n = 5 \times 10^5$ data points on an Intel(R) Xeon(R) Gold 6,252 CPU @ 2.10 GHz processor choosing $\alpha = 2$ and $\beta = 1$. As can be seen in Figure 4 (left), such run time is comparable to that of the `ocd R` package (Chen et al., 2022).

We also implemented Algorithm 1 for a change in the mean of a multi-dimensional Gaussian signal in R/C++ using the `qhull` library. We use the Gaussian distribution as an example of a

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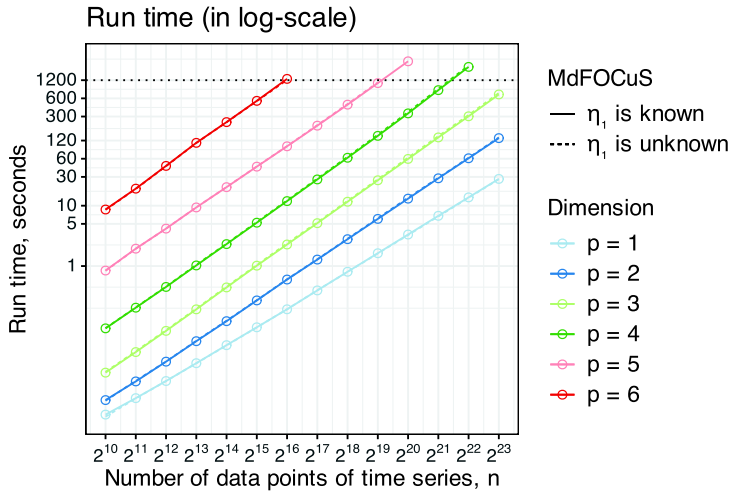


Figure 3. Run times in seconds of MdFOCuS with known (dashed line) and unknown (full line) pre-change parameter η_1, both with $\alpha = 2$ and $\beta = 1$, in dimension $p = 1, \ldots, 6$ for i.i.d Gaussian data. Run times are averaged over 100 data sets. We considered a maximum running time of 20 min (horizontal dotted black line) for the algorithms, and thus do not report run times for $p \geq 4$ for large n. The slope of the curves estimated are for known η_1: ≈ 1.005 ($p = 1$), ≈ 1.116 ($p = 2$), ≈ 1.180 ($p = 3$), ≈ 1.196 ($p = 4$), ≈ 1.143 ($p = 5$), and ≈ 1.202 ($p = 6$); and for unknown η_1: ≈ 1.006 ($p = 1$), ≈ 1.118 ($p = 2$), ≈ 1.183 ($p = 3$), ≈ 1.202 ($p = 4$), ≈ 1.144 ($p = 5$), and ≈ 1.197 ($p = 6$).

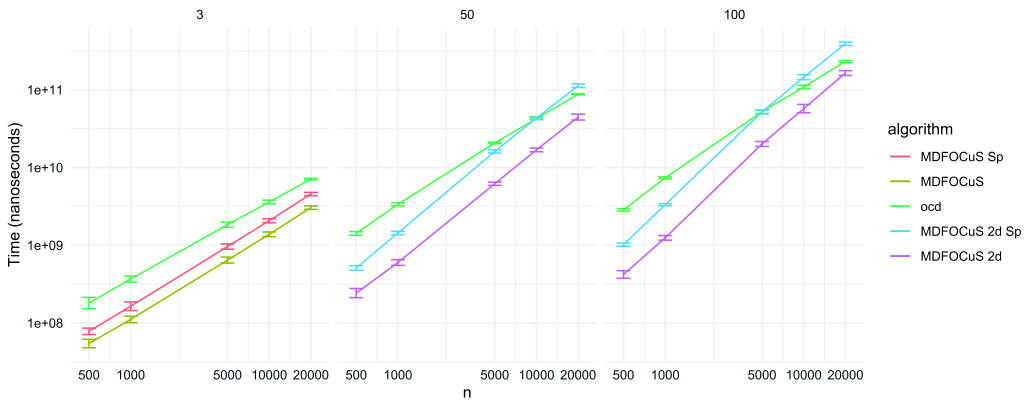


Figure 4. Average run times of MdFOCuS, MdFOCuS with sparse statistics (Sp), and the 2d hull approximation and `ocd` over 50 replicates for each scenario, for $p = 3$, 50, and 100 and n varying from 500 to 10,000 on an Intel Xeon Gold 6,342 (32) @ 2.793 GHz.

distribution with continuous density and independent dimensions. To be specific, for p from 1 to 6, we simulated time series $x_{1:n}$, with $x_t \sim \mathcal{N}_p(0, I_p)$ i.i.d. We varied the time series length from $2^{10} + 1 (\approx 10^3)$ to $2^{23} + 1 (\approx 8 \times 10^6)$ and simulated 100 data sets for each length. The average run times as a function of n are presented on the log-scale in Figure 3. As expected, it is faster than our `R` implementation: for example, for $p = 3$ it processes in 2 min for $n = 10^6$ rather than 5×10^5. The run times for a known or unknown η_1 are almost identical. Finally, we made a simple linear regression to model the logarithm of the run time as a function of $\log(n)$. The estimated slope coefficients are reported in Figure 3 and are all smaller than 1.2.

4.4 A provably quasi-linear online implementation

We showed how Algorithm 1 is exact and empirically fast, but it is non-trivial to derive its worst or expected time complexity as the time at which we run `QuickHull` is stochastic. In online

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supplementary material G, we propose Algorithm 3 that is closely related but easier to analyse. This algorithm updates the set $\mathcal{T}$ in a deterministic manner on predetermined chunks of the observations. This deterministic choice of chunks allows for a precise quantification of the expected complexity of the algorithm using Corollary 1 and the assumption that the convex hull algorithm is at worst quadratic. To be specific, here are our assumptions.

Assumption 1 In any dimension $p \geq 1$, there are positive constants c_1 and c_2 for which the expected number of vertices, $\mathbb{E}[V_n^p]$, can be bounded as follows:

$$\mathbb{E}[V_n^p] \leq c_1 \log(n)^p + c_2.$$

This is true under the conditions of Theorem 6 and Corollary 1. Next, define $Time(n)$ to be the worst-case computational cost of finding the convex hull of n points.

Assumption 2 For our convex hull algorithm there exist two positive constants c_3 and c_4 such that $Time(n) \leq c_3 n^2 + c_4$.

The assumption on the quadratic complexity is true for `QuickHull` for $p \leq 2$ (Barber et al., 1996) and using the algorithm of Chazelle (1993) it is true for $p \leq 3$.

In essence, Algorithm 3 computes the convex hull using a divide-and-conquer strategy. Out of simplicity, let us consider the offline setting and assume $n-1$ is a power of 2, that is $n-1 = 2^{\log_2(n-1)}$. Rather than running the hull algorithm on all points $\{P(\tau)\}_{\tau \in \{1,\ldots,n-1\}}$ we first run the algorithm on smaller chunks of size $2^{q_{\min}} : \tau \in \{2^{q_{\min}} j + 1, \ldots, 2^{q_{\min}} j + 2^{q_{\min}}\}$, with q_{min} a constant chosen by the user and j a number smaller than $(n-1)/2^{q_{\min}}$, and then dyadically merge the results. To be specific for each scale q with $q_{\min} < q < \log_2(n-1)$, we get the points on chunk $\{2^{q+1} j + 1, \ldots, 2^{q+1} j + 2^{q+1}\}$, by merging the results of the two smaller chunks $\{2^{q+1} j + 1, \ldots, 2^{q+1} j + 2^q\}$, and $\{2^{q+1} j + 2^q + 1, \ldots, 2^{q+1} j + 2^{q+1}\}$. This merging is computationally efficient because, under Assumption 1, the number of points on the hull of a chunk of size 2^q is expected to be small (of order $(\log(2^q))^p = (q \log(2))^p$).

One difficulty in Algorithm 3 is to make this dyadic or divide-and-conquer merging in an online fashion. To the best of our efforts, this makes the formal description a bit tedious, and we refer the interested reader to the online supplementary material G and Algorithm 2 and 3 for a precise description. We now state our bound on the expected complexity of online supplementary material Algorithms 2 and 3.

Lemma 1 Under Assumptions 1 and 2, the expected time complexity of Algorithm 2 and 3 is $\mathcal{O}(n \log(n)^{p+1})$.

We empirically study the complexity of Algorithm 3 in Figure 16 of online supplementary material G, where we see that it is slower than Algorithm 1. It is interesting to compare the bound on the computational cost with that of an oracle algorithm that knows the set of points on the convex hull at each iteration and whose computational cost is only evaluating the test statistic for each point on the hull. This would have a complexity of order $n \log^p(n)$, as $\sum_{t=1}^n \log^p(t) \geq n/2 \log^p(n/2)$. This is just a factor of $\log n$ faster than the bound in Lemma 1.

4.5 Two variant of MdFOCuS

In this subsection, we describe two variants of the algorithm, arising from variations in (i) the test statistic construction and (ii) the convex hull update. These variants are available regardless of whether or not the pre-change parameter is known.

The first variant will allow MdFOCuS to improve on scenarios where only a subset of the coordinates change: out of simplicity in lines 7 and 8 of Algorithm 1 we only consider the likelihood ratio statistics (5) that assumes all coordinates are changing. However, based on Theorem 5, the changepoints maximizing the sparse likelihood ratio statistics (2), where only s coordinates are changing, also correspond to points on the hull, and we can compute the test statistics in this case by replacing lines 7 and 8 with the corresponding sparse likelihoods.

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To compute all sparse max likelihood ratio statistics, we sort the per-dimension ratio statistics (this can be done in $\mathcal{O}(p \log(p))$) and sum these from the largest to the smallest in $\mathcal{O}(p)$ to iteratively recover the best for all sparsity levels s. We then recover the maximum for each sparsity level in $\mathcal{O}(\mathcal{T})$. If we consider all sparsity levels, the total complexity is $\mathcal{O}(|\mathcal{T}|p \log(p))$.

In practice, it makes sense to consider a subset of all sparsity levels. In our simulations we consider: $s \in \{2^0, 2^1, 2^2, \ldots, 2^{\lfloor \log_2(p) \rfloor}, p\}$. In a similar fashion to the FOCuS approximation, we also consider the sum of the max likelihood ratio statistics for each coordinate. As can be seen in Figure 4, the empirical run time of MdFOCuS with sparse statistics is comparable to that of `ocd` (Chen et al., 2022).

The second variant extends the algorithm to high-dimensional settings, where a full convex hull update is computationally prohibitive. In fact, under the condition of Theorem 6, Algorithm 1 is expected to store $\mathcal{O}(\log^p(n))$ candidates and the $\log^p(n)$ term means that the algorithm is impractical for p larger than 5, as exemplified in Figure 3. In online supplementary material H, we propose an approximate algorithm by recovering only a subset of the points on the hull. We obtain this subset by considering projections on at most $\tilde{p}$ variates, running the hull algorithm on each projection, and taking the union of the time-points (of the vertices) retained for each projection. For each projection or group of variates, under the condition of Theorem 6, we store $\mathcal{O}(\log^{\tilde{p}}(n))$ candidates. Considering $O(p)$ such groups we store $\mathcal{O}(p \log^{\tilde{p}}(n))$ candidates, thus reducing the computational complexity.

This heuristic for $\tilde{p} = 2$, implemented in `R`, runs on average in about 100 s for $p = 100$ and $n = 10{,}000$. We further explore the run times of this heuristic in Figure 4 and show that its run time is comparable to `ocd`. As shown in Section 4.7, it provides good statistical performance.

4.6 Statistical guarantees

In online supplementary material I, we provide several statistical guarantees for our procedure considering sparse or dense statistics in the multivariate Gaussian case. Our results also cover test statistics that threshold the contribution from each component. We consider properties of our algorithm if there is no change and then if there is one. In the first scenario, we cover both the pre-change parameter being known or unknown, while in the second, we only consider the pre-change parameter being known. We summarize the main results and their implications here.

We first consider the properties of our algorithm if there is no change. In this case, one may want to control either the false-positive rate, that is, the probability of ever detecting a change, or the average run length, the expected time until we detect a change. Our results give thresholds for both cases, but we will summarize them here just for an average run length of γ. At time n, we define the test statistic assuming a sparsity level s and a candidate change at τ as $S^s_{\tau,n}$. Testing for m different sparsity levels, and with the following threshold for sparsity s

$$c_s(\gamma) = s + 2\zeta + 2\sqrt{s\zeta}, \quad \text{where } \zeta = 4\log\gamma + s\log p + \log m + 5\log 2$$

our MdFOCus algorithm will have an average run length of at least γ (this follows by applying the argument in online supplementary material Remark 4 to the result of Proposition 8 with $N = 2\gamma$ and $\alpha = 1/2$). In the case where $s = p$, the $s\log(p)$ term in ζ can be omitted.

Next, we consider the power of detecting a change. This can be quantified in terms of the average detection delay, informally the expected time after a change that you would detect it, or bounds on the detection delay, the probability of detecting a change within some time-interval that it occurs. We provide results on both but will summarize here only the latter. Let the change in mean be specified by the vector δ. For a test at sparsity level s, denote by $\delta^{[s]}$ the vector of length s that consists of the s entries of δ with the largest absolute value. Finally, let τ^* denote the true changepoint and T^s the time at which we detect a change using the s-sparse statistic. Then from online supplementary material Proposition 9 and Remark 7, and using the threshold $c_s(\gamma)$, we have that with probability at least $(1-\alpha)$,

$$T^s - \tau^* < \frac{2(c_s(\gamma) - s) + 2\sqrt{s\log(3/\alpha)} - 8\log(\alpha/3)}{\|\delta^{[s]}\|^2}. \tag{11}$$

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We can draw some insights from these results. Excluding $s = p$, first observe that $c_s(\gamma)$ is increasing with s. This means, unsurprisingly, that to minimize the bound on the detection delay $T^s - \tau^*$, we never want to use s greater than the true sparsity of δ, with the notable exception of $s = p$. More generally, it is helpful to consider two different regimes depending on the relative sizes of γ and p. The first is a low-dimensional setting where $\log\gamma$ is much larger than p. In this case, the numerator of the right-hand side of (11) is dominated by the $\log\gamma$ term in $c_s(\gamma)$, and will be approximately $8\log\gamma/\|\delta^{[s]}\|^2$. In this case, maximizing $\|\delta^{[s]}\|^2$ is key to minimizing the detection delay, and this can be guaranteed by using the dense test statistic, $s = p$. Any advantage of using smaller s, in the case that some components do not change, will be small.

The other regime is where p is much larger than $\log\gamma$. In this case, using the dense statistics will give a detection delay with leading order (from the right-hand side of Equation (11)) that is proportional to $\sqrt{p}/\|\delta\|^2$. By comparison, the s-sparse statistic will have a detection delay with leading order proportional to $(s\log p)/\|\delta^{[s]}\|^2$. If only s components change, and all change by the same amount, then this latter quantity will be lower order in p when $s = o(\sqrt{p})$, as is consistent with other results on high-dimensional change detection (e.g. Chen et al., 2022). Finally, we can see that the benefit of robustness of performance to the true level of sparsity that comes using $m = O(\log p)$ tests for differing sparsity levels is only a $\log\log p$ increase in the thresholds and hence in the detection delays.

4.7 Empirical statistical performance

In this section, we evaluate the empirical performance of the `R` implementation of MdFOCuS with the sparse statistics (as in Section 4.5) on synthetic 3, 5, and 100-dimensional time series under the Gaussian change-in-mean setting, with an additional simulation study for a 50-dimensional time series presented in online supplementary material F.3. We consider two scenarios: pre-change mean known and unknown. For each experiment, we report results aggregated from 300 data sets.

For MdFOCuS for dimension $p = 3$ and 5, we consider sparsity level $s = 1$, the full max-likelihood ratio (either $s = 3$ or $s = 5$ depending on the dimension), and the sum of the max likelihood ratio statistics per dimension as proposed in Romano et al. (2023). For higher dimensions ($p = 100$ and 50), we use the approximate algorithm with $\tilde{p} = 2$ that analyses $2d$-projections of the data and set the sparsity levels as in Section 4.5. We compare the performance of MdFOCuS with the high-dimensional online changepoint detection method `ocd` (Chen et al., 2022) and a naive multivariate implementation of FOCuS, obtained by running FOCuS independently for each dimension and summing or taking the maximum of the resulting traces (see Section 6 of Romano et al., 2023) at each iteration. Additionally, as a comparison, for the pre-change mean unknown case, we report results of `ocd` and the pre-change mean known FOCuS with a plug-in estimate of the pre-change parameter, obtained over a probation (training) period of 250 observations.

Thresholds were selected under a Monte Carlo simulation to achieve an average run length of 5,000 observations. An additional study controlling the false positive rate can be found in online supplementary material F.3. For a change at $t = 1{,}000$, we evaluate performances for a range of different change magnitudes and sparsity levels. To compare performances across different sparsity regimes, changes in the affected time series were normalized such that the change direction lies in the p-dimensional sphere, i.e. ($\|\eta_1 - \eta_2\|_2^2 = \delta$), where δ is the overall change magnitude and η_1, η_2 are the pre- and post-change means, respectively. Tables 1 to 3 show results of the Average Detection Delays (ADDs) for 3, 5, and 100-dimensional time series, respectively.

For all dimensions of time series, in both pre-change parameter known and unknown scenarios, we observe that MdFOCuS generally achieves smaller detection delays in cases where the change affects all the time series and over changes of larger magnitudes. In scenarios with sparsity set to 1, the simple FOCuS aggregation tends to be slightly better. This is expected as if we only have a change in one dimension, both the sparse statistics of FOCuS and MdFOCuS will be the same, but MdFOCuS will have a slightly larger threshold, resulting in a slower delay.

For the pre-change mean known case, over changes of smaller magnitudes, we find MdFOCuS to be relatively close to the best competitor in terms of ADD, while it generally outperforms the other methods for larger changes. Furthermore, MdFOCuS is seen to have consistent performances across different sparsity levels. As `ocd` uses a grid for the possible size of change for each variate, its performance is affected by how close the true change size is to the nearest point on the grid.

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Table 1. Average detection delay with 3-dimensional time series for a change at time $t = 1{,}000$

Magnitude	Sparsity	Pre-change known			Pre-change unknown			
		FOCuS	MdFOCuS	ocd	FOCuS (est 250)	FOCuS	MdFOCuS	ocd (est 250)
0.125	1	**786.67**	806.54	1,227.57	**1,470.76**	1,644.05	1,726.24	1,783.41
0.125	2	**791.36**	817.12	1,421.54	**1,693.06**	1,770.33	1,815.88	1,987.04
0.125	3	**859.03**	886.83	1,496.18	**1,723.76**	1,806.46	1,853.35	2,097.23
0.250	1	250.58	255.78	**227.18**	470.69	**340.46**	364.50	430.33
0.250	2	**253.56**	256.76	268.86	556.86	369.36	**363.55**	516.28
0.250	3	273.92	**266.27**	300.69	630.94	404.19	**395.40**	559.66
0.500	1	65.85	66.46	**56.46**	113.75	69.73	**69.36**	83.90
0.500	2	71.85	69.93	**65.18**	132.99	78.42	**75.28**	96.50
0.500	3	79.21	76.52	**74.50**	155.61	87.95	**84.11**	103.97
0.750	1	31.80	32.16	**28.95**	50.73	32.62	**33.22**	39.46
0.750	2	34.66	33.31	**32.52**	61.46	36.15	**35.35**	43.47
0.750	3	37.19	**34.84**	35.43	71.75	39.40	**36.47**	46.40
1.000	1	**18.51**	19.04	18.87	29.18	**19.06**	19.53	24.16
1.000	2	20.30	**19.53**	20.45	34.55	20.94	**20.42**	25.45
1.000	3	21.68	**20.27**	21.52	40.03	22.49	**20.72**	26.23
2.000	1	**5.35**	5.41	6.40	8.06	**5.43**	5.53	7.12
2.000	2	5.97	**5.72**	6.64	9.46	6.13	**5.78**	7.25
2.000	3	6.51	**6.09**	6.80	10.62	6.71	**6.16**	7.41

Note. Pre-change parameter is known on the left, and unknown on the right. Best results per row are highlighted in bold. For the pre-change unknown case FOCuS (est 250) and ocd estimate, the pre-change parameter from 250 training samples.

In particular, its performance deteriorates when the true change is outside the set of grid values. We observe this in its deterioration of performance for low magnitudes (0.125) and large magnitudes (1 or 2), in particular over dimensions $p = 3, 5$.

For the pre-change mean unknown scenario, we observe that MdFOCuS generally outperforms competitor methods for magnitudes larger than 0.250, with the exception of scenarios with a change in a single dimension, as explained above. Moreover, in dimensions $p = 5$ and 100, we can clearly see the advantages of performing the likelihood ratio test for a change when the pre-change mean is unknown over using a plug-in estimate for the pre-change parameter. The only exception occurs in the case of FOCuS, when estimating the pre-change parameter from the training data, for $p = 3$, which out-performs other methods for the smallest size of change. Whilst this might seem counter-intuitive, as the error from the estimation should propagate, in dimension $p = 3$ for a small signal-to-noise ratio, the estimation is good enough to not impair the average run length and increase the thresholds, while benefiting from an easier optimization (as we see in the pre-change mean known case). However, as we move to higher dimensions $p = 5$ and 100, it is unlikely to obtain a better estimate of the true pre-change parameter simultaneously in all dimensions. This results in larger thresholds to control the desired run length, and therefore in a global loss of power. Of course, having more training observations improves the results, in particular for $p = 3$, however, this still has little to no effect in higher dimensions (see online supplementary material F.4).

5 Application to NBA Plus–Minus data

In this section, we illustrate our methodology to detect a change in the Cleveland Cavaliers (an NBA team) Plus–Minus score from season 2010–2011 to 2017–2018. This example was proposed

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Table 2. Average detection delay with 5-dimensional time series for a change at time $t = 1{,}000$

Magnitude	Sparsity	Pre-change known			Pre-change unknown			
		FOCuS	MdFOCuS	ocd	FOCuS (est 250)	FOCuS	MdFOCuS	ocd (est 250)
0.125	1	**833.11**	866.74	911.17	2,219.97	**1,718.76**	1,802.03	2,458.66
0.125	2	**892.82**	918.78	1,106.68	2,221.24	**1,874.47**	2,049.54	2,441.70
0.125	3	**925.08**	940.54	1,101.38	2,255.65	**1,924.89**	2,036.49	2,493.92
0.125	4	**835.44**	836.87	1,059.06	2,102.15	**1,825.65**	1,881.13	2,296.26
0.125	5	953.85	**948.59**	1,297.37	2,228.96	**2,005.88**	2,062.98	2,448.01
0.250	1	259.75	267.12	**229.64**	871.35	**360.65**	368.47	1,064.11
0.250	2	281.77	279.88	**253.79**	963.88	**433.69**	450.35	1,084.21
0.250	3	297.41	289.83	**287.10**	1,004.43	**456.72**	467.69	1,077.31
0.250	4	260.23	**242.46**	265.40	918.63	413.22	**382.42**	852.33
0.250	5	304.13	**288.07**	309.16	1,023.24	504.75	**495.38**	1,000.05
0.500	1	72.98	73.81	**64.74**	206.83	**80.54**	80.69	247.08
0.500	2	84.75	80.83	**73.93**	243.50	95.86	**91.35**	253.56
0.500	3	91.31	84.41	**83.54**	267.82	103.57	**96.62**	252.69
0.500	4	80.89	**71.59**	75.94	250.83	94.23	**81.13**	199.82
0.500	5	95.41	**86.77**	89.48	293.72	110.49	**100.25**	235.87
0.750	1	33.44	34.02	**32.98**	91.82	**34.46**	35.00	108.72
0.750	2	39.21	37.79	**36.22**	113.10	40.34	**39.06**	113.74
0.750	3	41.29	**38.16**	38.61	124.88	44.27	**41.49**	110.91
0.750	4	37.03	**33.10**	34.71	116.62	41.61	**35.94**	87.09
0.750	5	45.65	**40.26**	42.24	137.55	50.88	**44.03**	104.84
1.000	1	**19.63**	19.84	21.37	51.75	**20.02**	20.30	62.33
1.000	2	22.88	**21.86**	22.88	63.86	23.55	**22.49**	62.99
1.000	3	24.67	**22.83**	23.65	71.35	25.38	**23.47**	62.24
1.000	4	22.45	**19.29**	20.32	67.41	23.76	**20.20**	49.88
1.000	5	27.18	**23.63**	25.07	81.02	28.62	**24.79**	61.01
2.000	1	**5.79**	5.84	7.25	13.40	**5.73**	5.82	15.78
2.000	2	6.88	**6.58**	7.47	16.57	6.86	**6.53**	15.66
2.000	3	7.21	**6.55**	7.33	18.62	7.29	**6.72**	15.74
2.000	4	6.64	**5.72**	6.10	17.92	6.92	**5.90**	12.99
2.000	5	7.91	**6.87**	7.40	21.30	8.24	**7.08**	15.55

Note. Pre-change parameter is known on the left, and unknown on the right. Best results per row are highlighted in bold. For the pre-change unknown case FOCuS (est 250) and ocd estimate, the pre-change parameter from 250 training samples.

in Shin et al. (2024). We use this example for two reasons. First, it highlights that our computational framework is also applicable to CUSUM Exponential baseline e-detectors proposed in Shin et al. (2024) which in essence are a non parametric generalization of generalized likelihood ratio rules. Second, it illustrates how modelling not only the signal's mean but also its variance can simplify the calibration of the detection threshold, without necessarily resorting to a non-parametric approach.

We recovered the Plus–Minus score using the `nbastatR` package (Bresler, 2023) available at https://github.com/abresler/nbastatR. The Cavaliers Plus–Minus is represented in Figure 5 and is a time series of the difference in score of the Cavaliers and their opponents over successive games. We consider detecting a changepoint in this data online.

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Table 3. Average detection delay with 100-dimensional time series for a change at time $t = 1{,}000$

Magnitude	Sparsity	Pre-change known			Pre-change unknown			
		FOCuS	MdFOCuS	ocd	FOCuS (est 250)	FOCuS	MdFOCuS	ocd (est 250)
0.125	1	1,417.15	1,391.54	**1,171.63**	3,459.08	2,708.45	**2,704.65**	3,546.43
0.125	5	1,881.10	1,810.57	**1,627.00**	3,497.94	**2,859.51**	2,872.20	3,595.24
0.125	10	2,084.84	1,978.85	**1,940.41**	3,501.48	**2,846.45**	2,874.17	3,595.46
0.125	50	**2,252.48**	2,289.56	2,386.98	3,494.67	2,888.46	**2,881.59**	3,591.96
0.125	100	**2,323.59**	2,338.00	2,431.51	3,507.66	**2,859.11**	2,887.60	3,599.24
0.250	1	389.11	380.09	**335.95**	2,659.50	730.54	**720.45**	2,841.37
0.250	5	641.18	555.57	**505.39**	3,054.36	1,520.80	**1,451.53**	3,232.02
0.250	10	789.27	673.15	**595.10**	3,178.29	1,879.64	**1,692.20**	3,334.21
0.250	50	1,057.30	895.53	**827.30**	3,323.17	2,353.01	**2,309.73**	3,453.27
0.250	100	1,080.26	916.64	**891.09**	3,346.50	**2,341.24**	2,414.38	3,476.58
0.500	1	105.79	107.89	**104.79**	721.53	**118.84**	121.50	859.71
0.500	5	169.93	147.36	**139.23**	1,297.63	214.14	**185.07**	1,449.56
0.500	10	232.04	176.44	**170.03**	1,653.97	296.51	**217.84**	1,850.22
0.500	50	357.90	248.41	**239.08**	2,480.88	578.17	**348.83**	2,719.69
0.500	100	372.34	254.51	**244.07**	2,611.86	679.19	**365.79**	2,841.33
0.750	1	**47.78**	48.67	57.56	310.89	**50.08**	50.68	461.51
0.750	5	75.34	**66.80**	73.47	580.05	85.34	**73.90**	715.73
0.750	10	104.45	**80.81**	84.16	798.60	117.25	**88.30**	925.56
0.750	50	184.09	116.47	**114.29**	1,626.21	239.73	**133.60**	1,836.28
0.750	100	193.77	120.24	**115.21**	1,823.30	279.42	**137.98**	2,074.96
1.000	1	**27.12**	27.55	38.67	173.04	**27.76**	28.41	317.49
1.000	5	43.72	**39.02**	46.07	325.28	46.44	**41.45**	476.20
1.000	10	60.36	**47.40**	50.77	458.62	63.85	**49.35**	599.74
1.000	50	109.11	67.33	**66.02**	1,074.41	135.79	**73.06**	1,218.93
1.000	100	116.60	69.23	**67.59**	1,288.45	157.47	**75.55**	1,466.28
2.000	1	**7.58**	7.65	12.93	43.03	**7.53**	7.73	142.00
2.000	5	12.92	**11.12**	13.80	82.41	12.95	**11.33**	203.48
2.000	10	16.46	**13.16**	14.80	117.67	16.30	**13.48**	250.44
2.000	50	31.85	18.37	**18.08**	315.56	34.85	**18.70**	451.59
2.000	100	34.28	18.97	**18.34**	422.46	41.93	**19.37**	493.39

Note. Pre-change parameter is known on the left, and unknown on the right. Best results per row are highlighted in bold. For the pre-change unknown case FOCuS (est 250) and ocd estimate, the pre-change parameter from 250 training samples.

As detailed in Shin et al. (2024), it is nontrivial to fit this problem into commonly used parametric sequential changepoint detection procedures for two main reasons. First, it is not easy to choose a parametric model to fit the score. In particular, we would like to take into account that we have integer-valued data, and we expect some form of dependency. Second, assuming we have chosen a model or likelihood function, calibrating the threshold to detect a changepoint is not trivial. A small change in the distribution we use as a reference can substantially change the threshold and as a consequence increase the detection time or lead to false detections. E-detectors (Shin et al., 2024), are a non-parametric solution to these two problems.

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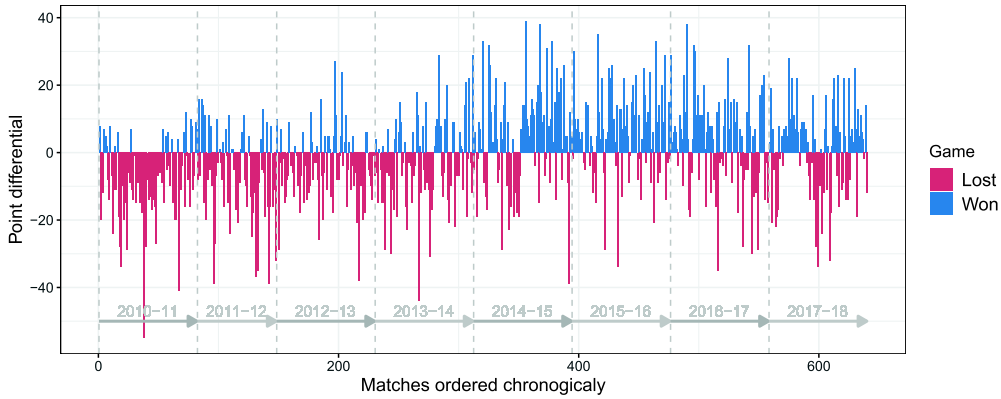


Figure 5. Plus–Minus score of the Cavaliers from season 2010–2011 to season 2017–2018 as a bar plot. Losses are downward bars and wins are upward bars. Ends of seasons are represented with vertical dashed lines and seasons are shown on the bottom with horizontal arrows.

5.1 Convex hull for CUSUM e-detectors

Following Equations (27) and (41) of Shin et al. (2024), we can write their update of the e-detector CUSUM statistics for a given λ as

$$M_n^{CU}(\lambda) = \exp\left(\lambda s(x_n) - \psi(\lambda)v(x_n)\right) \max\left\{M_{n-1}^{CU}(\lambda),\, 1\right\},$$

where λ is in a subset of $\mathbb{R}$, s and v are real-valued functions, and ψ is a finite and strictly convex function. Taking logs and considering all possible changepoints prior to n we recover with induction that

$$\log\left(M_n^{CU}(\lambda)\right) = \max_{\tau<n}\left\{\sum_{t=\tau+1}^{n}\left(\lambda s(x_t) - \psi(\lambda)v(x_t)\right)\right\}.$$

By subtracting $\sum_{t=1}^{n}\left(s(x_t)\lambda - \psi(\lambda)v(x_t)\right)$ in the previous bracket, we do not change the maximizer, and we recover our functional pruning framework of Equation (8) with $a_\tau = \sum_{t=1}^{\tau} v(x_t)$, and $2b_\tau = \sum_{t=1}^{\tau} s(x_t)$. Applying Theorem 4, we get that the changepoint optimizing the e-detector CUSUM score for any λ value corresponds to a vertex on the convex hull of

$$\left(\sum_{t=1}^{\tau} v(x_t),\ \frac{1}{2}\sum_{t=1}^{\tau} s(x_t)\right)_{\tau<n}. \tag{12}$$

As argued in Shin et al. (2024), it makes sense to consider a mixture of e-detectors, that is different λ values. For computational reasons, Shin et al. (2024) considered a finite number of values, chosen using their `computeBaseLine` function.

We will now compare the baseline numbers of two e-detectors for Plus–Minus score to the number of points on the hull of $eq : edetect - hull(12)$. Shin et al. (2024) considered two e-detector models. In the first, which we call Winning rate, they model the winning rate and consider $v(x_t) = 1$ and $s(x_t) = \mathbb{1}_{x_t>0} - p_0$, where $\mathbb{1}$ is the indicator function and p_0 is taken to be 0.49 in the numerical application. With `computeBaseLine` they selected 69 λ values. In their second e-detector model, which we call Plus–Minus, they analyse the Plus–Minus score more directly. The score is first normalized between 0 and 1 as $x'_t = (x_t + 80)/160$. Then they define $s(x') = (x'/m - 1)$ and $v(x) = (x'/m - 1)^2$, where m is chosen to be 0.494 in the numerical application. With `computeBaseLine` they selected 190 λ values. For both e-detectors, we computed the number of points in the convex hull for n between 10 and $n = 640$ (the last match of season 2017–2018)

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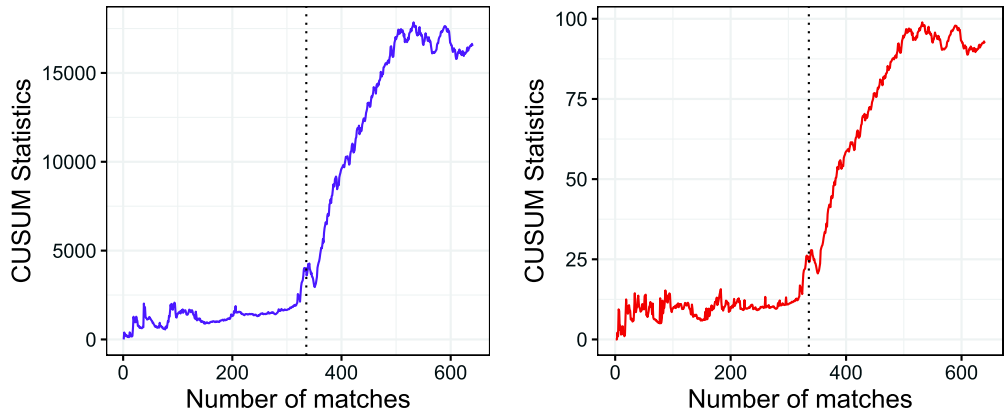


Figure 6. CUSUM statistics for the mean model (left) for the mean and variance model (right). The dotted line corresponds to the beginning of season 2014–2015 and match number 312.

and obtained for Winning rate at most 16 and for Plus–Minus at most 18 points. Therefore out of the, respectively, 69 and 190 candidates suggested by `computeBaseLine` many are redundant as they correspond to the same changepoint.

5.2 A parametric approach to the Plus–Minus score

Considering a change in the mean of a Gaussian model to fit the Plus–Minus score of the Cavaliers does not take into account the discrete nature of the data and possible dependencies. It is also not clear how one could get a valid estimation of the pre-changepoint parameter. Finally, as argued in Shin et al. (2024) calibrating the threshold to detect a changepoint is not trivial. As a robust parametric solution to these problems, we consider a change in the mean and variance of a univariate Gaussian signal with unknown pre-changepoint parameters.

For a change in mean and variance using our convex hull framework (8), we need to keep track of the vertices on the hull of a 3-dimensional set of points

$$\left(\tau, \sum_{t=1}^{\tau} x_t, \sum_{t=1}^{\tau} x_t^2\right)_{\tau<n}. \tag{13}$$

Interestingly, this mean and variance model is invariant to shift and scaling. Based on that, we see that all the points on the convex hull of the Plus–Minus e-detector are on the hull of the change in mean and variance model.

As the data are discrete, two (or more) consecutive scores may be exactly equal. For segments containing equal scores, the estimated variance would be 0, and the likelihood would blow up to infinity. To get around this problem, we considered a minimum value for the variance, which we set to 1 (chosen as substantially below the variance seen for Plus–Minus statistics for NBA teams).

In Figure 6, we represent the CUSUM statistics for the change in mean and change in mean and variance model. Although the scales are quite different, the two statistics have very similar trends and importantly both seem to increase around season 2013–2014 (starting at match number 312 represented by a vertical dotted line). Although both models reconstruct the same changepoint, we will see in the next subsection that the mean-variance model shows less variability in terms of detection delay, achieving faster detections once a threshold is set to achieve a fixed run length.

Interestingly, at any time-point after 320 (8th match of season 2014–2015) the maximum likelihood changepoint is always between matches 279 and 280 (5th and 7th of February 2014) for both models (see Figure 7). This is well before the beginning of season 2014–2015 and the comeback of Lebron James. However, this matches the arrival of David Griffin as general manager on the 6th of February (Wikipedia, 2023).

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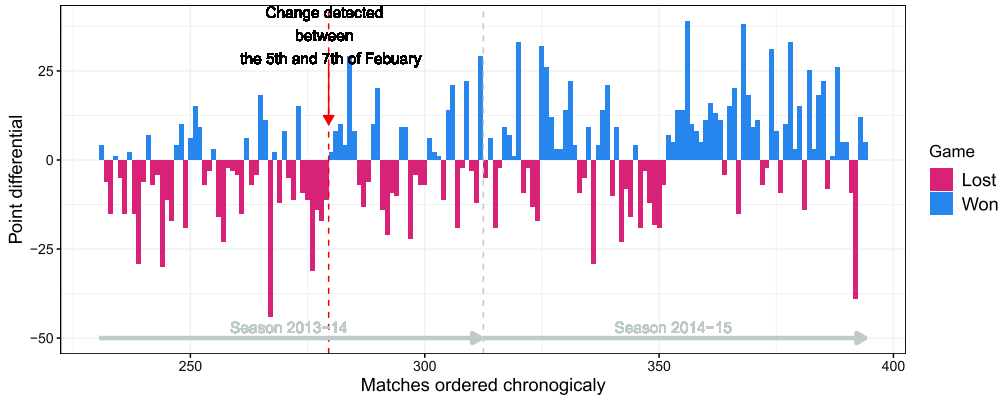


Figure 7. Plus–Minus score of the Cavaliers from seasons 2013–2014 to 2014–2015 as a bar plot. Losses are downward bars and wins are upward bars. The ends of seasons are represented with vertical dashed lines and seasons are shown on the bottom with horizontal arrows.

5.3 Simulating NBA Plus–Minus scores without changepoint

Based on Figure 6, it is clear that for a particular range of the threshold both the mean and mean and variance model with unknown pre-changepoint parameter can detect a changepoint after the beginning of season 2014–2015. To calibrate this threshold, one typically needs a simulation model. Below, we consider different choices of simulation model to see how sensitive the threshold is to this choice for both models. In all cases, we will set the CUSUM threshold such that under our null simulation over 1,000 matches we would detect a change only in 5% of the cases.

The simulation models we considered are: (i) the difference of two independent Poisson random variables; (ii) the difference of two negative binomial random variables; (iii) a uniform distribution on $\{-80, \ldots, 80\}$ (as suggested in Shin et al., 2024); (iv) an independent resampling of past games. For (i) and (ii), we consider a range of parameters, with the mean of the random variables set to be 95, 105, or 115; and with the negative-binomial scale parameter set to 1 or 2. For (iv), we considered either sampling past matches of the Cavaliers or of NBA teams in general, from seasons 1999/2000 to 2009/2010. Finally, we also considered version of the scenarios in (i) and (ii) where we introduced temporal dependencies by adding to the mean a time-varying function either simulated from an AR(1) model with auto-correlation of 0.6 or using a sine function (Figure 8).

Figure 9 shows the various thresholds we get for our two models, and the corresponding detection times. The threshold is much more variable for the change in mean model and this translates to a much more variable detection time. To be specific for the mean model, the change is detected at the earliest at match 330 (the 5th of December 2014), but in 21.7% of the cases the threshold is so high that the changepoint is never detected (reported in the histogram as a detection at 1,000). For the mean and variance model, the change is always detected at the earliest at match 339 (the 23rd of December 2014) and at the latest at match 368 (the 20th of February 2015). In fact in 94.2% of the simulation models, the change is detected before with the mean and variance model.

6 Discussion

In this article, we establish a connection between the detection of a single changepoint in p-dimensional data streams and a convex hull problem in dimension $p + 1$. Based on this, we derive a simple yet efficient algorithm, MdFOCuS. This algorithm finds a set of time-points that contain the points associated with all vertices of a convex hull, and that we show contains all possible times that can maximize the likelihood-ratio statistic for a change. The method is general in that in can be used for a wide-range of models from the exponential family, including multivariate data where different streams are modelled by different distributions. It can also deal with the pre-change parameter being known or unknown, and covers tests where we allow only a subset of

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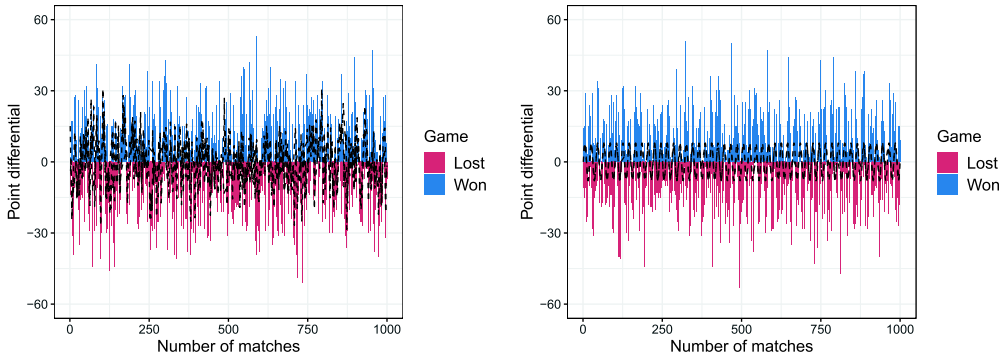


Figure 8. Two Plus–Minus score simulation examples based on the negative binomial model with a mean of 115 and scale parameter of 50 using two dependency scenarios: an AR(1) model with AR parameter 0.6 (left) and using sine waves with a period of 20 matches (right).

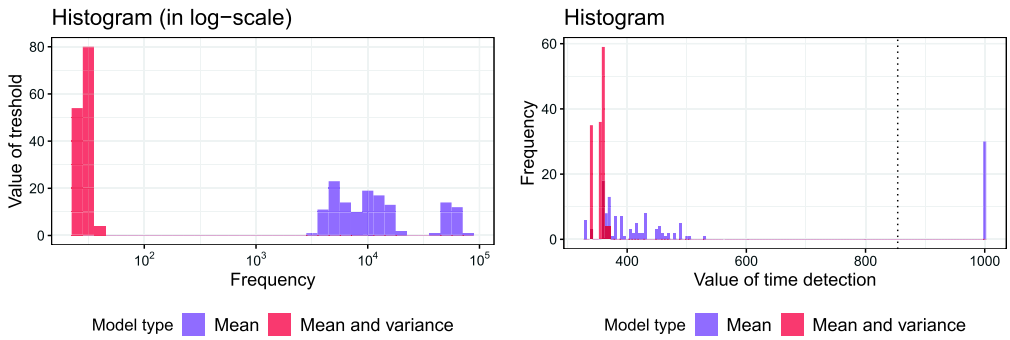


Figure 9. Threshold (left) and detection time (right) obtained for the mean model and the mean and variance model.

components of a multivariate data stream change. For all these varieties, the only difference is the part of the algorithm which maximizes the function associated with each time-point we have stored. We also showed that it can be used to speed up the calculation of e-detectors. MdFOCuS is efficient for $p \leq 5$. For larger p, we provide an approximation storing only $\mathcal{O}(p \log^{\tilde{p}}(n))$ vertices of the convex hull where $\tilde{p} \leq p$.

Improvements to the computational complexity of the procedure could be made to allow the procedure to scale to higher dimensions and longer sequences. For example, a sequential update of the boundary of the convex hull of our points, to improve on the complexity of `QuickHull`. Recently, Ward et al. (2024) offered an approach to speed-up the maximization step. This aims to recycle calculations to give a bound on the test statistic that is much more efficient to calculate. Whilst this idea can directly be applied in the multivariate settings we consider, the bounds are too loose to give any noticeable speed-up. A less conservative bound could be the object of future study.

More generally, the ideas in this article which show a connection between changepoint detection and convex hull (or half-space intersection) problems sheds light on the geometric nature of the changepoint detection problem and is a promising line for future research.

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Conflicts of interest: None declared.

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Data availability

We recovered the Plus–Minus score using the nbastatR package (Bresler, 2023) available at https://github.com/abresler/nbastatR.

Supplementary material

Supplementary material is available online at *Journal of the Royal Statistical Society: Series B*.

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